Blood types and inheritance

Science 9, Module 2: Heredity and variation. Read this handout before the next class.

The ABO blood groups

A person's ABO blood type depends on the markers, called antigens, found on the surface of their red blood cells. There are four ABO blood types: A, B, AB and O. Type A blood has the A antigen, type B has the B antigen, type AB has both, and type O has neither.

One gene with three alleles

The ABO blood type is controlled by one gene that has three alleles: I^A, I^B and i. Each person carries two of these alleles, one from each parent. This is an example of multiple alleles, because the gene has more than two forms in the population.

The alleles I^A and I^B are codominant. When a person has both, both antigens appear and the blood type is AB. Both I^A and I^B are dominant over i, so i only shows when a person has two copies of it.

Genotype	Blood type
I^AI^A or I^Ai	A
I^BI^B or I^Bi	B
I^AI^B	AB
ii	O

Example 1: a type AB parent and a type O parent

The type AB parent has the genotype I^AI^B. The type O parent has the genotype ii. Each child gets I^A or I^B from the first parent and i from the second. Half of the children are expected to be I^Ai (type A) and half I^Bi (type B). None of their children can have type AB or type O blood.

Example 2: two type A parents who both carry i

Both parents have the genotype I^Ai. A Punnett square gives 1 I^AI^A : 2 I^Ai : 1 ii. So three out of four children are expected to have type A blood, and one out of four is expected to have type O blood, even though both parents are type A.

The Rh factor

Blood is also either Rh positive or Rh negative, depending on another antigen called the Rh factor. Rh positive is dominant over Rh negative. A person whose blood is "O positive" has type O blood and is Rh positive.

Why blood type matters

Before a blood transfusion, the blood types of the donor and the patient must be checked. If the patient's body sees the donor's antigens as foreign, it attacks the donated red blood cells. People with type O negative blood are often called universal donors, because their red blood cells carry no A, B or Rh antigen. People with type AB positive blood are often called universal recipients.

Try it

One parent has the genotype I^Ai and the other has I^Bi. What blood types can their children have? Draw the Punnett square to check your answer.